C6: Video Analysis [MO83391]

Ball Action Spotting

Team 2:

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Agenda

Júlia

01 Introduction

02 Dataset

03 Summary of previous weeks

04 Our Best Model for Action Spotting

05 Ablations

06 Summary

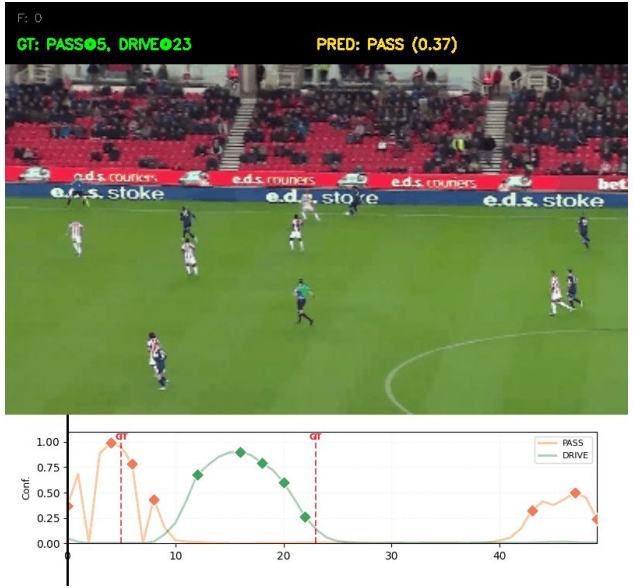
07 Insights & Discussion

08 Conclusions & Future Work

Introduction

Júlia

Action Spotting



SoccernetBall Dataset

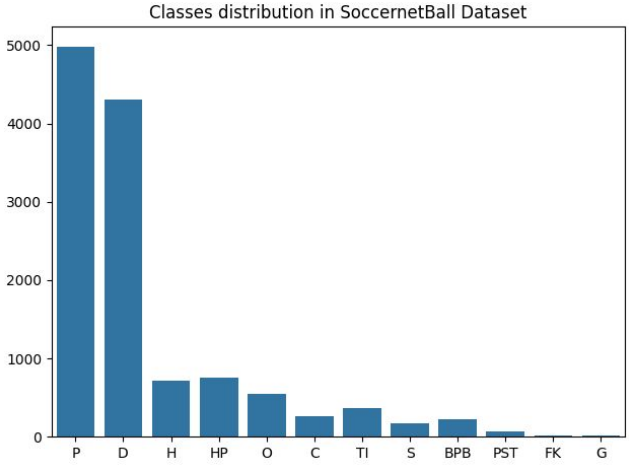
Júlia

Info

Total: 720p 25FPS
Train: 4 videos | 576.615 frames
Val: 1 Video | 146.892 frames
Test: 2 Videos | 298.653 frames

Classes

PASS	OUT	BALL PLAYER BLOCK
DRIVE	CROSS	PLAYER SUCCESFUL TACKLE
HEADER	THROW IN	FREE KICK
HIGH PASS	SHOT	GOAL



Summary of previous weeks

Júlia

Without reducing the input's temporal resolution

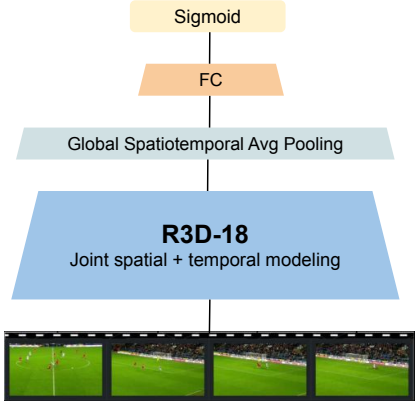
Week 5

Ball Action Classification

Best Model: R3D-18 + CLIP LEN 100 + STRIDE 5

+32,19% w.r.t W5's baseline

	AP10	Num. Params	MACs (G)
Best W5	60,37	33.172.428	70,74



Clip (L = 100 frames)

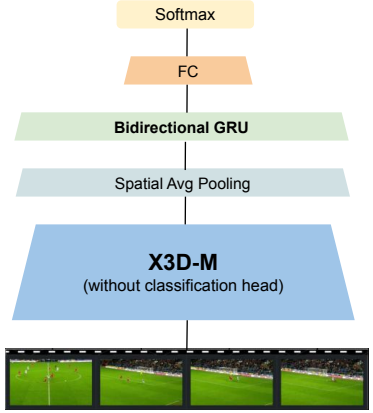
Week 6

Ball Action Spotting

Best Model: X3D-M + BIDIRECTIONAL GRU

+40% w.r.t W6's baseline

	AP10	Num. Params	MACs (G)
Best W6	44,34	4.243.951	28,077



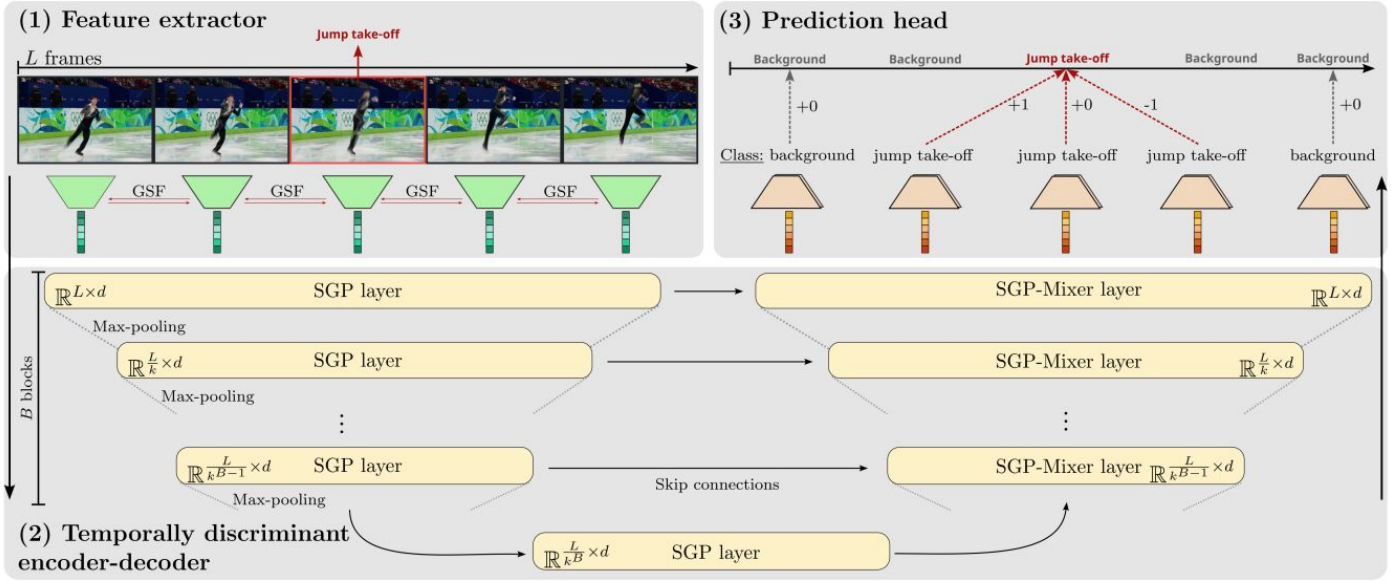
Clip (L = 50 frames)

T-DEED: A UNet Like Approach

Lore

Key Ideas

Multi-scale temporal modeling (encoder–decoder)	Frame-level predictions for precise localization	Temporal refinement via displacement prediction	Efficient design (2D backbone + lightweight temporal modules)



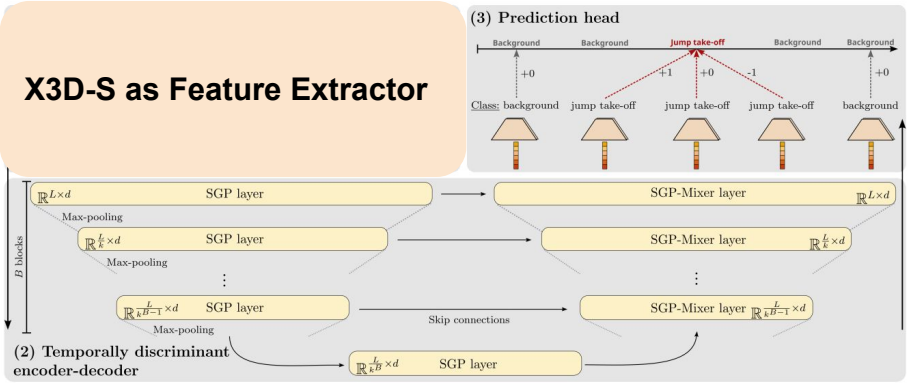
Our best model for Action Spotting

Lore

UNet-Like approach

Week 7
Ball Action Spotting
Best Model: T-DEED (X3D-S) + DISPLACEMENT (r=2)

X3D-S as Feature Extractor



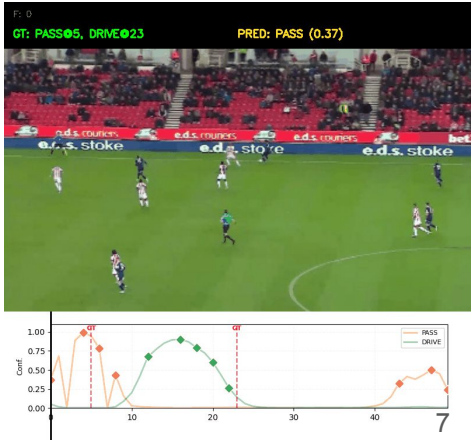
Hyperparameters:

- Radius displacement: 2
- N. of temporal layers: 2
- SGP kernel size: 9
- SGP scaling factor: 4
- Crop dim: -1

Class	AP10 (vs. Baseline)
PASS	79,40 (+61,68)
DRIVE	68,96 (+54,40)
HEADER	50,00 (+48,60)
HIGH PASS	69,13 (+60,04)
OUT	19,43 (+19,04)
CROSS	61,34 (+61,10)
THROW IN	56,88 (+56,88)
SHOT	39,38 (+39,38)
BALL PLAYER BLOCK	15,97 (+15,97)
PLAYER SUCCESSFUL TACKLE	0,12 (+0,12)
FREE KICK	36,36 (+36,36)
GOAL	3,03 (+3,03)

	Mean AP10 (vs. Baseline)
AP 10	**46,06 (+41,72)**
AP 12	**41,67 (+37,33)**

	Best Model
MACs (G)	28,145
Num. Params	6.440.624



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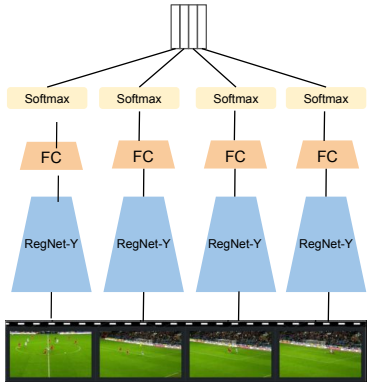
Ablations - Baseline

Lore

Class	AP10
PASS	17,72
DRIVE	14,56
HEADER	1,4
HIGH PASS	9,09
OUT	0,39
CROSS	0,24
THROW IN	0,00
SHOT	0,00
BALL PLAYER BLOCK	0,00
PLAYER SUCCESSFUL TACKLE	0,00
FREE KICK	0,00
GOAL	0,00

	Mean AP10
AP 10	4,34
AP 12	3,62

	Baseline
MACs	18,105
Num. Params	2.798.793



Clip (L = 50 frames)

Settings

FIXED

- **Clip Length**: 50
- **Stride**: 2
- **Batch Size**: 4
- **Num. Frames/epoch**: 500000
- **LR**: 0,0008
- **Num. Epochs**: 20
- **Warm Up Epochs**: 3

NOT FIXED

- Early Stopping: No
- Tolerance: 1
- Backbone: RNY-002
- Neck: No
- Head: FC + Softmax

GPU: RTX 3090 / RTX 6000 ADA Generation

Ablations - U-Net Approach

Lore

T-DEED

Best (bold) | Second best (underlined)

Variation	AP10 (vs. Baseline)	Num. Params	MACs (G)
T-DEED (RNY-002)*	35,67 (+31,33)	12.291.561	18,631
T-DEED (RNY-008)*	34,18 (+29,84)	46.185.237	73,303
T-DEED (X3D-S)	44,22 (+39,88)	6.440.431	28,145
T-DEED (X3D-M)	43,98 (+39,64)	6.440.431	28,145
T-DEED (X3D-L)	43,17 (+38,83)	8.799.541	54,451
X3D M + GRU-BIDIR	**44,34 (+40,00)**	**4.243.951**	**28,077**
RNY-002 + LSTM-BIDIR	29,17 (+24,83)	4.976.249	18,215

* With GSF

Settings

- Early Stopping: Yes
- Patience: 5 Epochs
- Tolerance: 1

- Temporal Arch.: "ed_sgp_mixer"
- Radi Displacement: 0
- Num. Layers: 2
- SGP KS: 9
- SGP R: 4
- CROP DIM: -1 (No cropping)

Insights

- Not much change from last week
- Pretrained and Finetuned X3D is the main impact
- Pyramid architecture may be better than RNN

Ablations - U-Net Approach

Lore

T-DEED

Best (bold) | Second best (underlined)

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- Radi Displacement: 0
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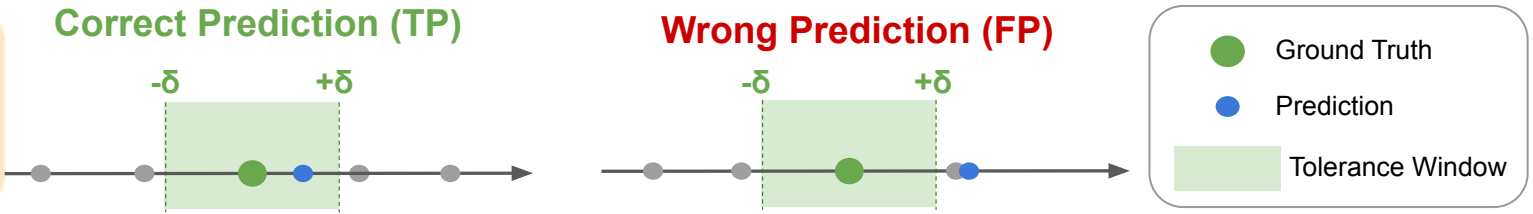
Insights

- Not much change from last week
- Pretrained and Finetuned X3D is the main impact
- Pyramid architecture may be better than RNN

Ablations - Tolerance

Adrián

Tolerance (δ)
Maximum allowed temporal distance between prediction and GT (seconds).



Ablation Setup	Question	Hypothesis
Evaluate multiple tolerances: coarser (**1s**) and tighter (**0.5s**) Perform a fair comparison between a full temporal resolution model and a temporal aggregation model	Does temporal aggregation harm **precision**?	Temporal aggregation may reduce temporal precision due to information compression Its impact is expected to be more visible under tighter tolerance (0.5 s) than coarser tolerance (1.0 s)

Ablations - Tolerance

Adrián

Tolerance

Best (bold)

Variation	AP10 (δ=0.5)	AP10 (δ=1)	Num. Params	MACs (G)
T-DEED (X3D-S)	38,92	44,22	**6.440.431**	**28,145**
X3D-L + GRU-BIDIR	**39,03**	**44,88**	6.603.061	54,383

Insights

- AP10 increases for both models at **1.0 s tolerance**
- Temporal aggregation has **minimal impact on performance**
- Gap between models is **similar across tolerances**
- Suggests **temporal precision is largely preserved**

Settings

COMMON

- Early Stopping: Yes
- Patience: 5 Epochs
- Tolerance: 0.5/1
- Head: FC + Softmax

W/ TEMPORAL AGGREGATION

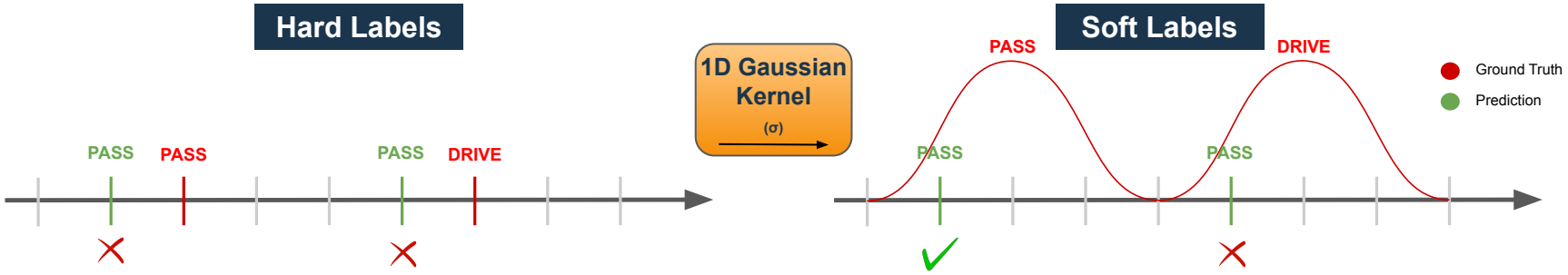
- Backbone: X3D-S
- Temporal Arch.: "ed_sgp_mixer"
- Radi Displacement: 0
- Num. Layers: 2
- SGP KS: 9
- SGP R: 4
- CROP DIM: -1 (No cropping)

W/O TEMPORAL AGGREGATION

- Backbone: X3D-L
- Temporal Arch.: Bidirectional GRU

Ablations - Soft Labels

Adrián



Ablation Setup

Compute loss using a gaussian distribution (soft labeling) over the true ground truth labels (hard labels)

Test different standard deviations (2, 4, 6, 12).

Window of gaussian computed as 2σ.

Question

Can this help the model by guiding it towards an approximation of the event instead of the exact frame?

Hypothesis

Soft labeling can help the model by accepting a prediction as correct with a reasonable offset of a few frames.

This could make the model more exact, but at the cost of being less precise.

Ablations - Soft Labels

Adrián

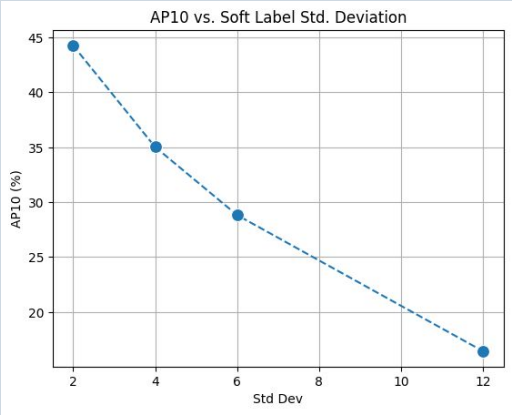
T-DEED (X3D-S)

Best (bold)

Std. Deviation	AP10 (vs. No Soft Label)
σ=2	**44,27** (+0,05)
σ=4	35,03 (-9,19)
σ=6	28,83 (-15,39)
σ=12	16,43 (-27,84)

Insights

- Adding Soft Labels does not help… what's going on?



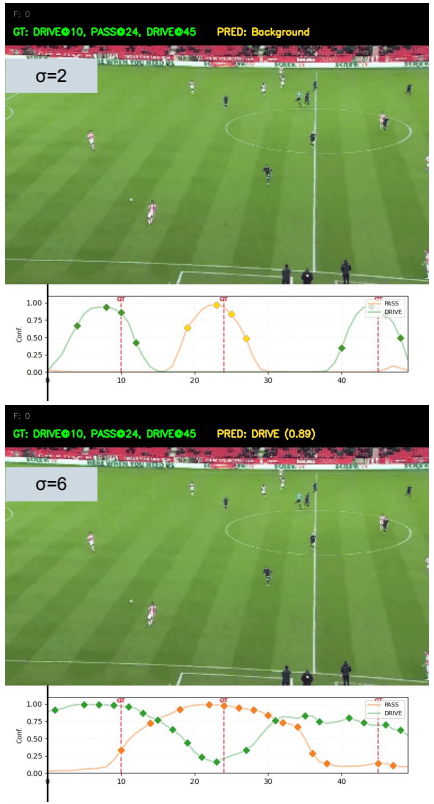
Settings

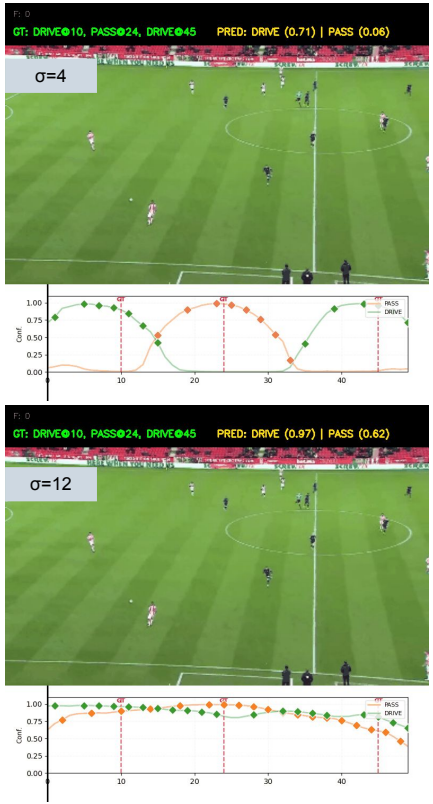
- Early Stopping: Yes
- Patience: 5 Epochs
- Tolerance: 1

- Temporal Arch.: "ed_sgp_mixer"
- Radi Displacement: 0
- Num. Layers: 2
- SGP KS: 9
- SGP R: 4
- CROP DIM: -1 (No cropping)

Ablations - Soft Labels

Adrián





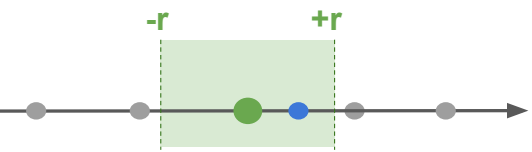
Insights

- Soft Labeling increases predictions (after NMS)
- Big σ → overlapping high confidence predictions
- Exactness/Precision trade-off not found

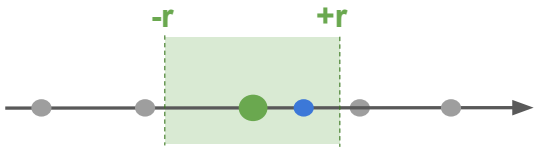
Ablations - Radius Displacement

Adrián

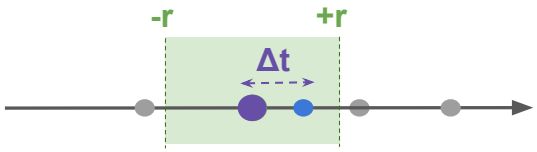
1) Detection Radius (r)

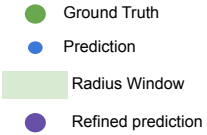


2) Coarse Prediction



3) Displacement Refinement





Ablation Setup	Question	Hypothesis
Evaluate the displacement head with different displacement radius (2,4) Compare performance relative to no displacement	What radius provides the best trade-off between easier detection and precise localization?	A small positive radius should improve performance by making near-event frames easier to detect and then refine A large radius may hurt precision, since too many frames are treated as positives and localization becomes ambiguous

Ablations - Radius Displacement

Adrián

T-DEED (X3D-S)

Best (bold)

Radius	AP10 (vs. R=0)
R=2	**46,06 (+1,84)**
R=4	44,26 (+0,04)

Insights

- Displacement improves performance
- Best results at **moderate radius (r = 2),** providing the best balance between detection and localization
- Larger radius offers **limited additional benefit**

Settings

- Early Stopping: Yes
- Patience: 5 Epochs
- Tolerance: 1

- Temporal Arch.: "ed_sgp_mixer"
- Radi Displacement: 2/4
- Num. Layers: 2
- SGP KS: 9
- SGP R: 4
- CROP DIM: -1 (No cropping)

Summary

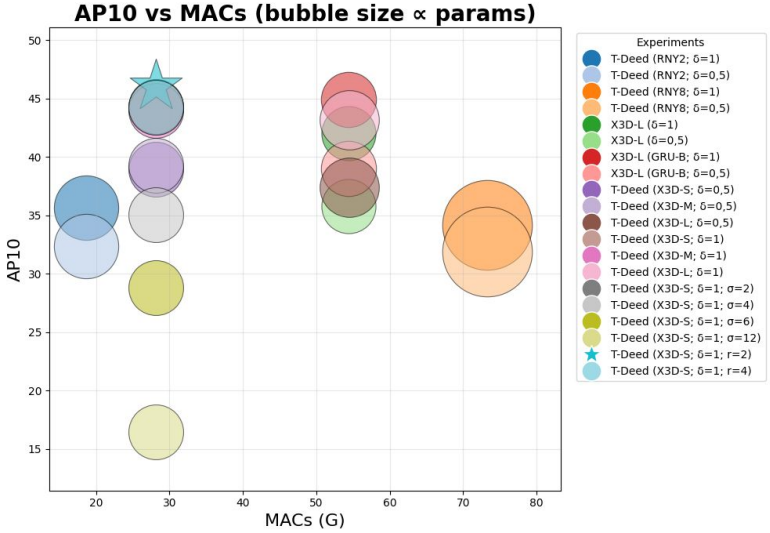
Marina

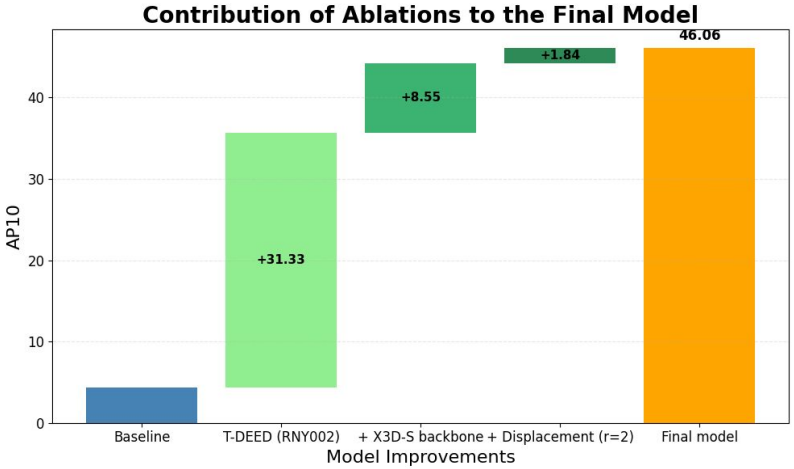
UNet-Like approach

Week 7

Ball Action Spotting

Best Model: T-DEED (X3D-S) + DISPLACEMENT (r=2)





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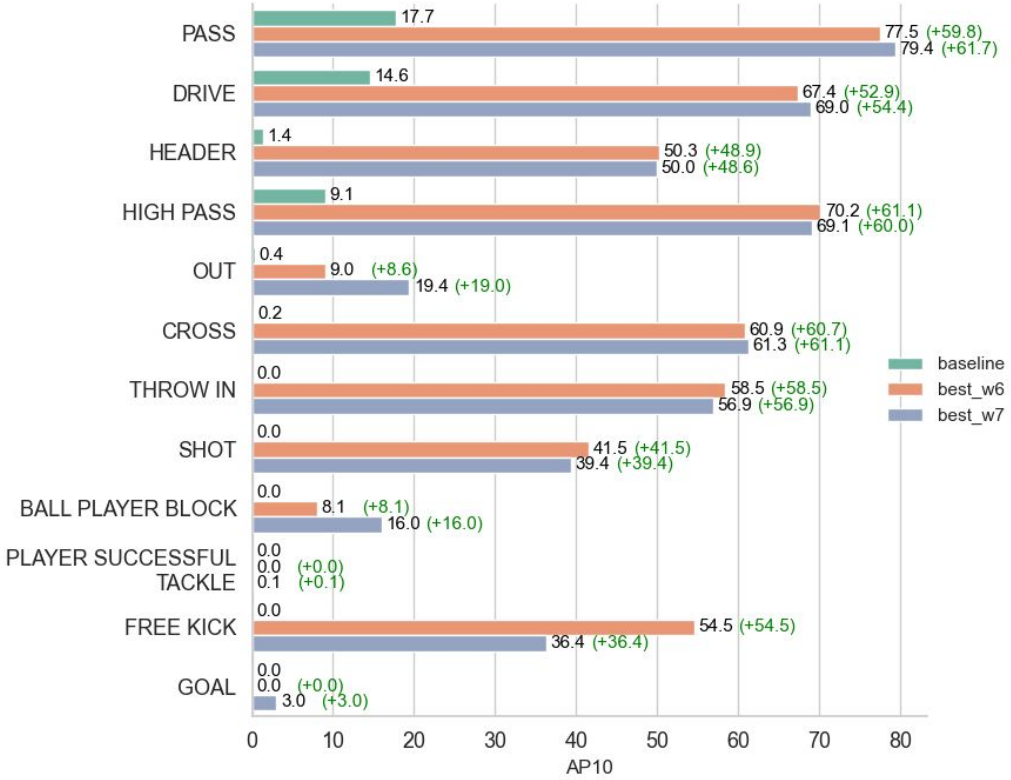
Insights & Discussion

Marina



	Mean AP10 (vs. Baseline)		
	Baseline	**Best W6**	**Best W7**
AP 10	4,34 (+0)	44,34 (+40,00)	**46,06 (+41,72)**
AP 12	3,62 (+0)	41,50 (+37,88)	**41,67 (+37,33)**

	Baseline	**Best W6**	**Best W7**
MACs (G)	28,145	28,077	28,145
Num. Params	6.440.624	4.243.951	6.440.624

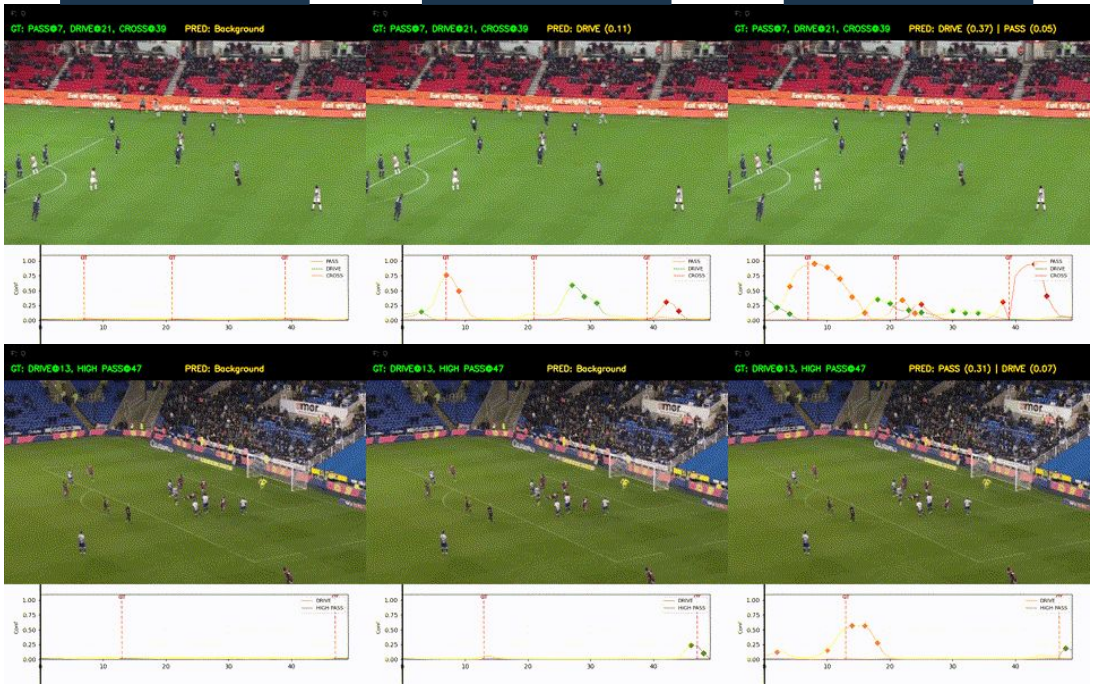
Insights

- Improvement of **+1,72%** w.r.t. W6's best model.
- Trade-off of **+2.196.480 (+51,76%) params** and **+0,068G (+00,24%) MACs** w.r.t. W6's best model.
- Temporal aggregation (U-Net-like) showed limited improvement over W6.
- Radius Displacement and X3D backbone had the most impact.
- Bottleneck: few and unbalanced data.
- Slight improvement in least represented classes.

Insights & Discussion

Marina

Baseline	Best W6	Best W7



Insights

- Clear improvement w.r.t. Baseline.
- More confidence in this week's predictions. (But more FP, even with NMS).
- W7 can become a bit chaotic (top example).
- Bottom: case of class improvement in W7 w.r.t. W6.

Conclusions & Future Work

Marina

Conclusions

- Best model: T-DEED (X3D-S) + DISPLACEMENT (r=2)
- Barely +1,72% AP10 increase w.r.t W6
- Ablations:
  - U-Net type
  - Tolerance impact
  - Soft labels
  - Radius Displacement

Future Work

- Change both tolerance and soft labels (strict NMS needed)
- Use a bigger context window (increased clip length)
- Finetune pretrained T-Deed (we have few data)

References

Marina

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- C. Feichtenhofer, “X3D: Expanding Architectures for Efficient Video Recognition,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
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- A. Xarles, S. Escalera, T. B. Moeslund and A. Clapés, “T-DEED: Temporal-Discriminability Enhancer Encoder-Decoder for Precise Event Spotting in Sports Videos,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 3410–3419.

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